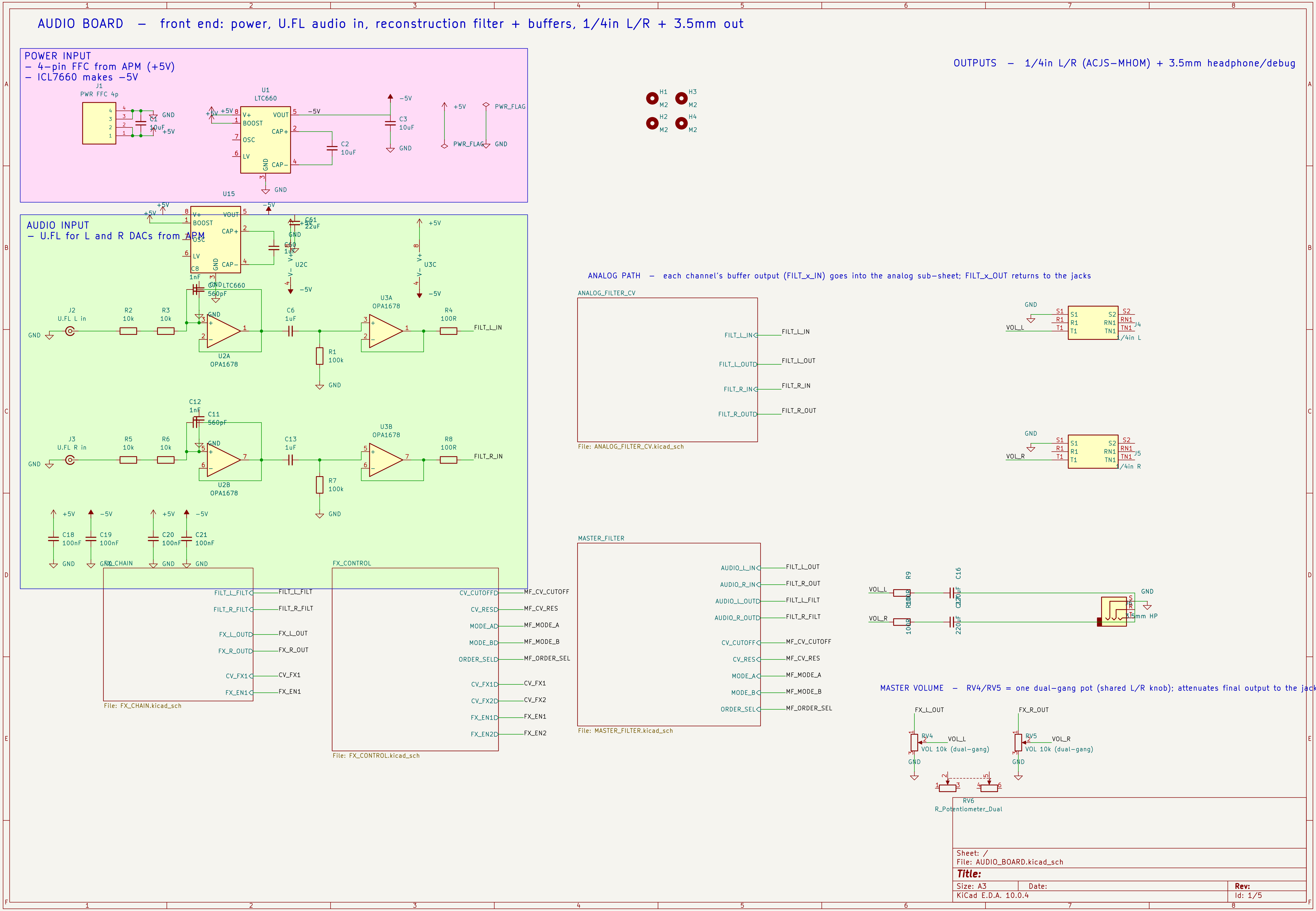
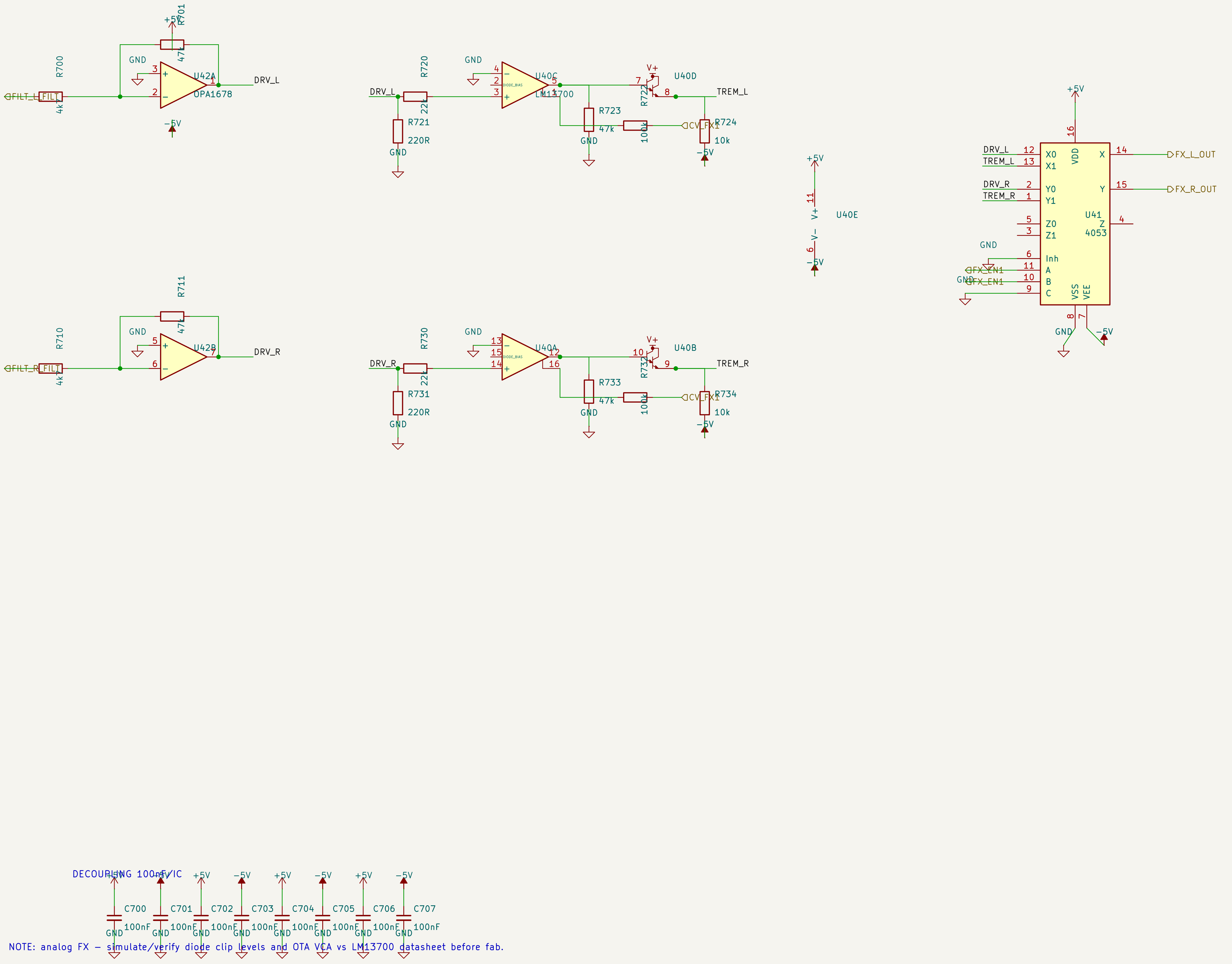

AUDIO BOARD – front end: power, U.FL audio in, reconstruction filter + buffers, 1/4in L/R + 3.5mm out
POWER INPUT
– 4-pin FFC from APM (+5V)
– ICL7660 makes –5V
J1
PWR FFC 4p
C1
10uF
GND
+5V
U1
LTC660
V+
BOOST
OSC
LV
GND
VOUT
CAP+
CAP–
–5V
C2
10uF
C3
10uF
PWR_FLAG
H1
H2
H3
H4
M2
OUTPUTS – 1/4in L/R (ACJS-MHOM) + 3.5mm headphone/debug
U15
LTC660
C61
22uF
C60
C8
1nF
C9
560pF
AUDIO INPUT
– U.FL for L and R DACs from APM
U2C
U3C
J2
U.FL L in
R2
10k
R3
10k
U2A
OPA1678
C6
1uF
R1
100k
U3A
OPA1678
R4
100R
FILT_L_IN
C12
1nF
C11
560pF
J3
U.FL R in
R5
10k
R6
10k
U2B
OPA1678
C13
1uF
R7
100k
U3B
OPA1678
R8
100R
FILT_R_IN
C18
100nF
C19
100nF
C20
100nF
C21
100nF
ANALOG PATH – each channel's buffer output (FILT_x_IN) goes into the analog sub-sheet; FILT_x_OUT returns to the jacks
ANALOG_FILTER_CV
FILT_L_IN
FILT_L_OUT
FILT_R_IN
FILT_R_OUT
File: ANALOG_FILTER_CV.kicad_sch
J4
1/4in L
J5
1/4in R
S1
R1
T1
S2
RN1
TN1
VOL_L
VOL_R
FX_CHAIN
FILT_L_FILT
FILT_R_FILT
FX_L_OUT
FX_R_OUT
CV_FX1
FX_EN1
File: FX_CHAIN.kicad_sch
FX_CONTROL
CV_CUTOFF
CV_RES
MODE_A
MODE_B
ORDER_SEL
CV_FX1
CV_FX2
FX_EN1
FX_EN2
MF_CV_CUTOFF
MF_CV_RES
MF_MODE_A
MF_MODE_B
MF_ORDER_SEL
File: FX_CONTROL.kicad_sch
MASTER_FILTER
AUDIO_L_IN
AUDIO_R_IN
AUDIO_L_OUT
AUDIO_R_OUT
File: MASTER_FILTER.kicad_sch
R9
100R
R10
100R
C16
220uF
C17
220uF
3.5mm HP
MASTER VOLUME – RV4/RV5 = one dual-gang pot (shared L/R knob); attenuates final output to the jacks/HP
RV4
VOL 10k (dual-gang)
RV5
VOL 10k (dual-gang)
RV6
R_Potentiometer_Dual
Sheet: /
File: AUDIO_BOARD.kicad_sch
Title:
Size: A3
Date:
KiCad E.D.A. 10.0.4
Rev:
Id: 1/5

FX CHAIN – DRIVE (soft clip) + TREMOLO (OTA VCA, LFO=CV_FX1); each bypassable (FX_EN). In: FILT_x_FILT (from SVF) -> Out: FX_x_OUT (to volume)




NOTE: analog FX – simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet before fab.

Sheet: /FX_CHAIN/
File: FX_CHAIN.kicad_sch

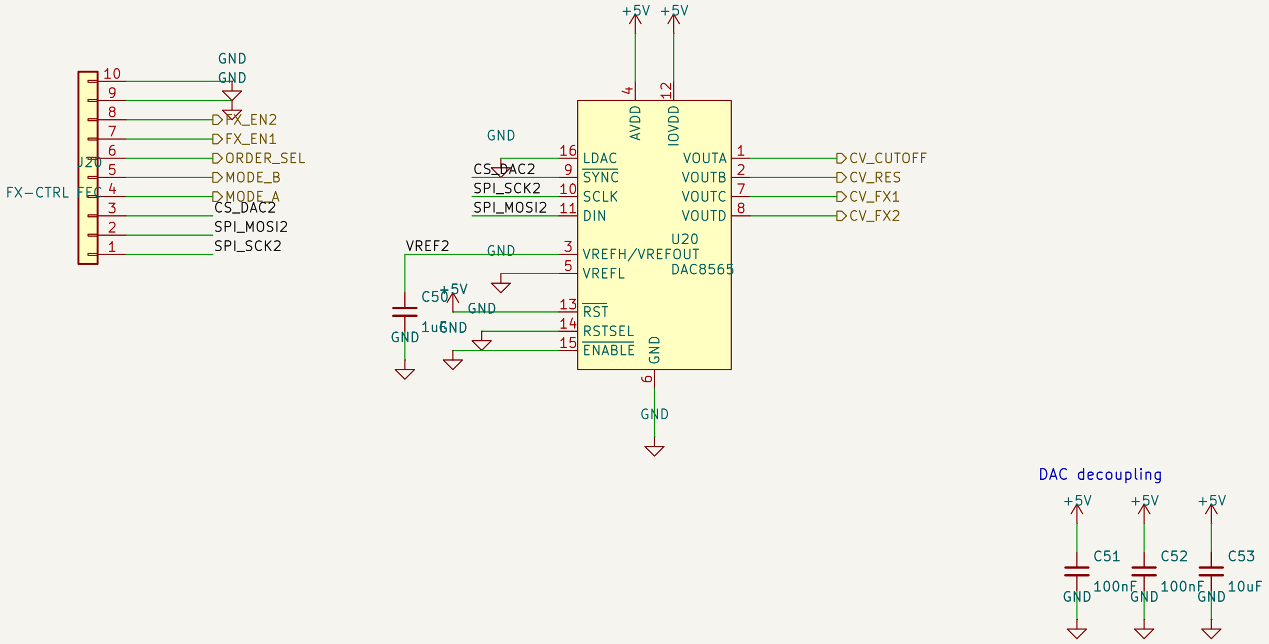
**Title:**

Size: A3 Date: Rev:
KiCad E.D.A. 10.0.4 Id: 5/5

FX CONTROL  –  2nd FX–control FFC (SPI + mode/order/enable) –> DAC8565 #2 makes CV_CUTOFF/CV_RES (+2 spare FX CVs); digital selects pass through

FX–CTRL FFC:  1SCK 2MOSI 3CS_DAC2 4MODE_A 5MODE_B 6ORDER_SEL 7FX_EN1 8FX_EN2 9/10GND

DAC8565 #2 –> 4 CV channels for the master filter/FX




outputs to filter/FX: CV_CUTOFF, CV_RES (–> SVF), CV_FX1/CV_FX2 (–> future phaser/chorus/tremolo); MODE_A/B, ORDER_SEL, FX_EN1/2 digital.

Knobs/switches live on the controller board; STM32 reads them, drives this DAC over SPI and sets the digital lines.

Sheet: /FX_CONTROL/
File: FX_CONTROL.kicad_sch

**Title:**

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| KiCad E.D.A. 10.0.4 | | Id: 3/5 |

ANALOG FILTER / CV (stereo) – SPI in (10-pin FFC) -> DAC8565 makes CVs -> per-channel LM13700 VCF/VCA; CD4053 selects analog or FPGA-digital

CONTROL – 10-pin FFC (SPI from APM)

DAC8565 – 4-ch SPI CV DAC (cutoff / res / VCA voltages, on-board)

TOGGLE – CD4053: MODE_SEL selects analog (VCA_x_OUT) vs bypass (FILT_x_IN) for BOTH L an

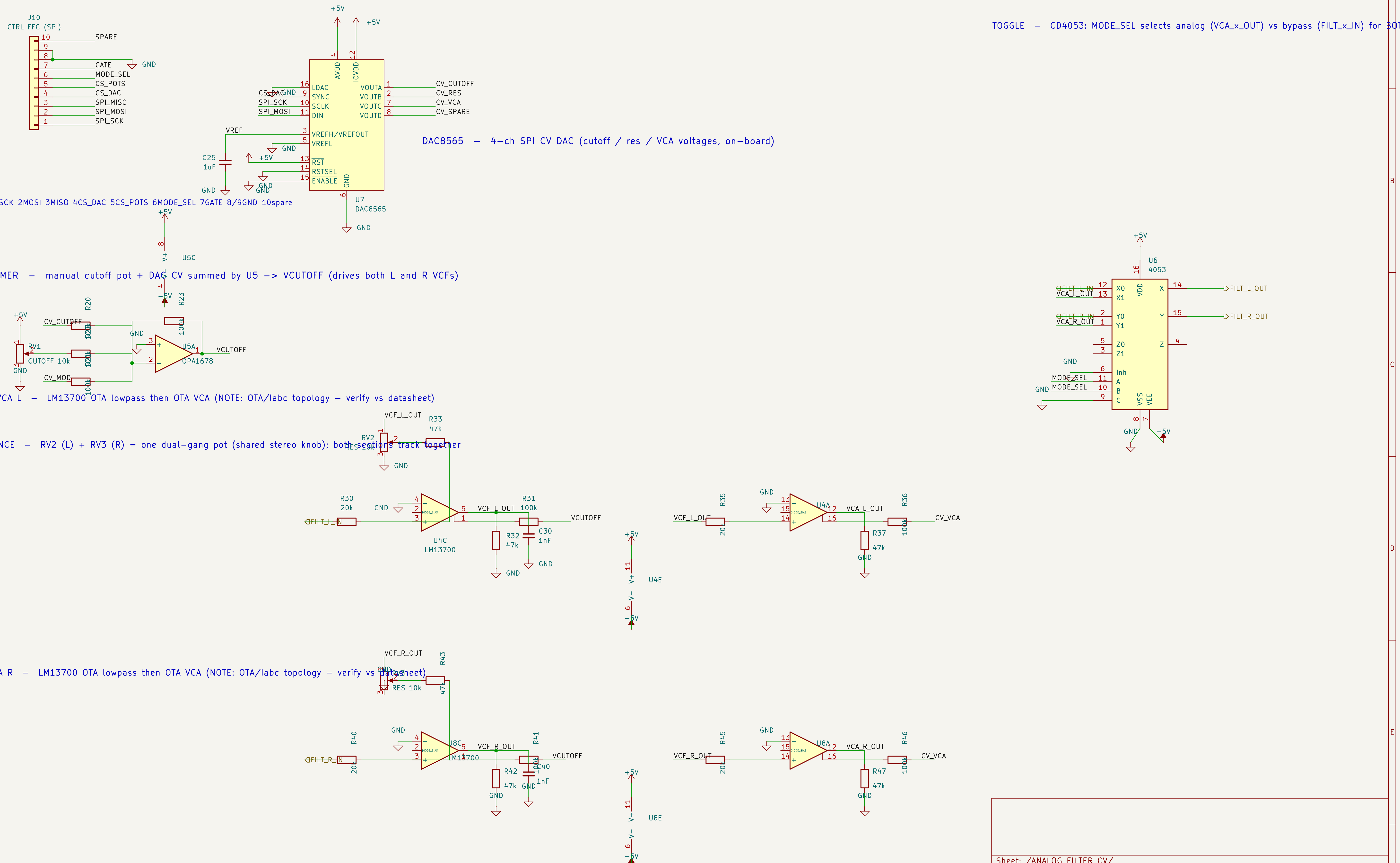
pinout: 1SCK 2MOSI 3MISO 4CS_DAC 5CS_POTS 6MODE_SEL 7GATE 8/9GND 10spare

CV SUMMER – manual cutoff pot + DAC CV summed by U5 -> VCUTOFF (drives both L and R VCFs)

VCF+VCA L – LM13700 OTA lowpass then OTA VCA (NOTE: OTA/Iabc topology – verify vs datasheet)

RESONANCE – RV2 (L) + RV3 (R) = one dual-gang pot (shared stereo knob); both sections track together

VCF+VCA R – LM13700 OTA lowpass then OTA VCA (NOTE: OTA/Iabc topology – verify vs datasheet)




FILT_x_IN / FILT_x_OUT are the hierarchical pins shared with the root (front-end) sheet. Power rails (+5V/-5V/GND) are global.

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**Title:**

| Size: A3 | Date: | Rev: |
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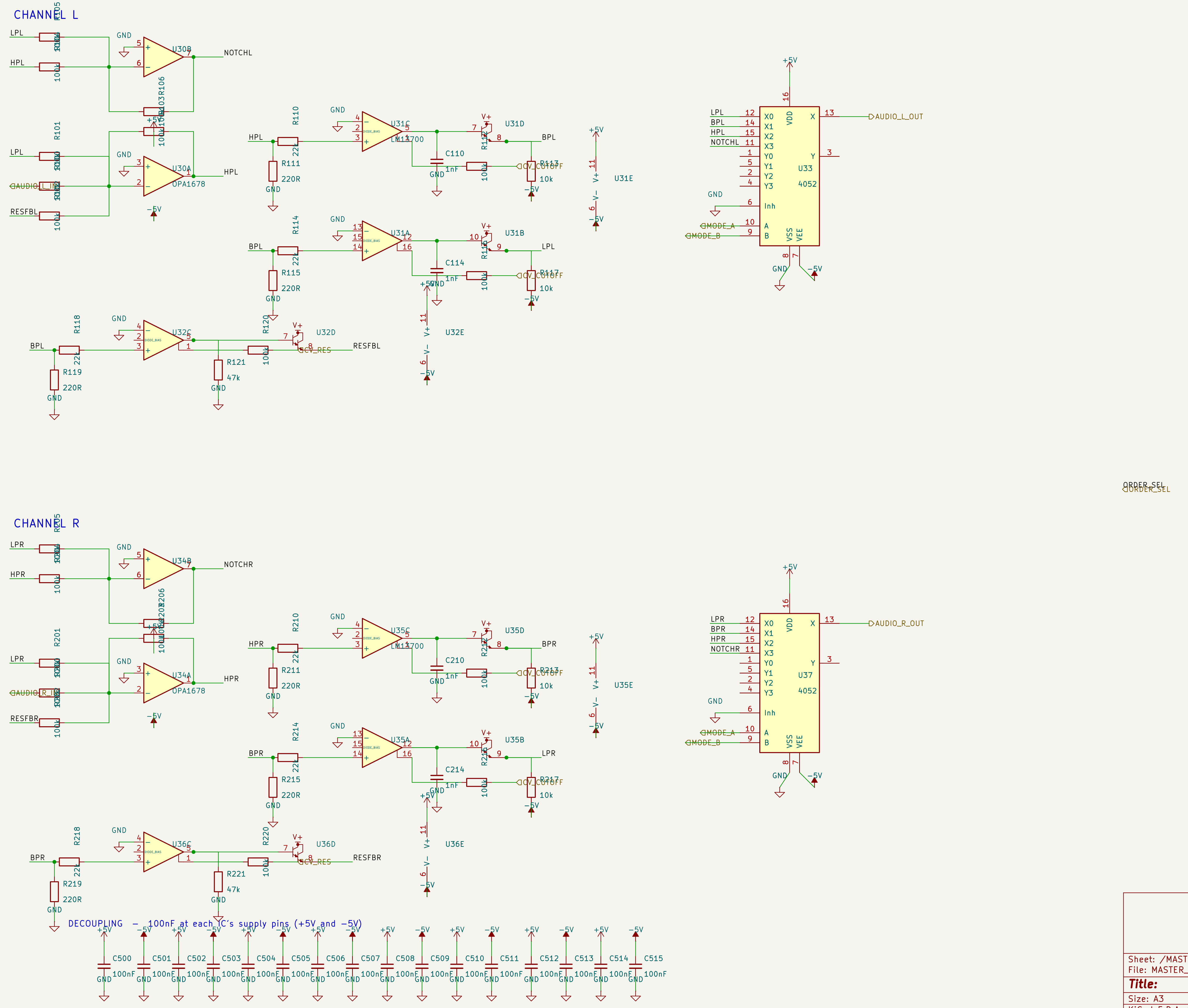
| KiCad E.D.A. 10.0.4 | | Id: 4/5 |

MASTER FILTER (stereo) – VC state-variable filter 2-pole: LP/HP/BP/notch; CV cutoff + CV resonance; mode-select mux

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File: MASTER_FILTER.kicad_sch

***Title:***

Size: A3 Date:

Rev:

KiCad E.D.A. 10.0.4

Id: 5/5

NOTE: OTA VC-SVF + OTA resonance VCA – prototype/verify levels, Iabc, buffer bias vs LM13700 datasheet before fab. ORDER_SEL reserved for the 2/4-pole cascade.